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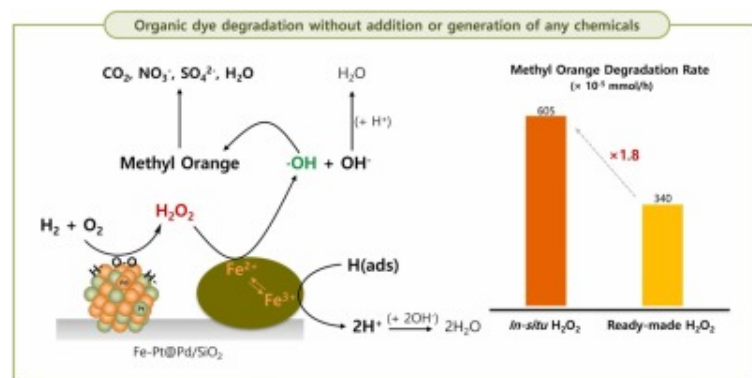
Highlights

- Catalytic wet peroxidation (CWPO) of methyl orange with *in-situ* generated H₂O₂ was proposed.
- Bifunctional Fe-Pt@Pd/SiO₂ catalyst prepared by double-metal complex salt approach was highly efficient.
- The CWPO reaction with *in-situ* generated H₂O₂ surpassed that with ready-made H₂O₂.
- The new approach enables wastewater treatment without any undesirable chemicals affecting environment.

Abstract

While the Fenton reaction is efficient in wastewater treatment, the intrinsic problem of the dosage-dependent process poses another burden on ecosystem. Herein, we propose a highly efficient and environmentally-benign catalytic wet peroxide oxidation (CWPO) of methyl orange (MO) using *in-situ* generated H₂O₂ from hydrogen and air instead of ready-made H₂O₂. Pt@Pd/SiO₂ and Pd@Pt/SiO₂ catalysts were prepared by double-metal complex salt approach, and they were combined with soluble Fe salt (CWPO-I) or impregnated Fe (CWPO-II). The MO degradation rate of Pt@Pd/SiO₂ for CWPO-I was 2.5 times faster than that of the conventional PdPt/SiO₂. This can be rationalized in terms of Pd-enriched surface with strong interaction between Pt and Pd. Moreover, bifunctional Fe-Pt@Pd/SiO₂ catalyst demonstrated outstanding activity in promoting CWPO-II with *in-situ* generated H₂O₂, even outperforming its counterpart with ready-made H₂O₂. This new approach offers an efficient water remediation method free from addition and generation of any undesirable chemicals affecting environment.

Graphical Abstract



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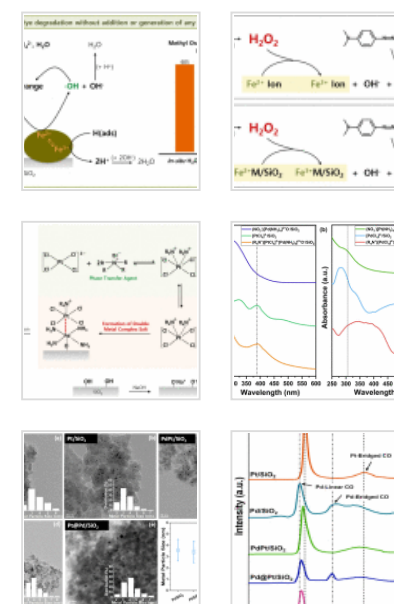
Appendix A. Supplementary material

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Catalytic wet peroxide oxidation; Wastewater treatment; Organic dye degradation; Direct synthesis of H₂O₂; Bifunctional catalyst

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1. Introduction

As urbanization and industrialization continue to advance, wastewater treatment has become an urgent and critical concern. Wastewater from various sources, including agriculture, municipalities, and industries, contains a host of organic and inorganic contaminants, as well as toxic compounds, all of which have a detrimental impact on humans, animals, and the environment [1]. Among these pollutants, synthetic dyes are particularly problematic in industrial wastewater. Annually, over 7×10^5 tons of synthetic dyes are used in industries like food, pharmaceuticals, textiles, printing, and chemicals [2], [3], [4]. Unfortunately, these dyes often contain potentially carcinogenic and non-biodegradable chemicals [5], [6], [7], [8]. Furthermore, azo dyes, which make up about 80% of synthetic dyes, have high solubility in water, making their removal a challenging task. In response to this, researchers have explored various strategies for efficiently removing or separating synthetic dyes from water, including the use of biological microorganisms, flotation, adsorption, membrane filtration, and thermal wet air oxidation. However, these methods have their limitations, such as high energy requirements, maintenance costs, and the need for additional treatments to prevent the formation of hazardous byproducts [9], [10], [11].

Recently, advanced oxidation processes (AOPs) have garnered attention. These processes involve the *in-situ* generation of highly reactive hydroxyl or sulfate radicals through the decomposition of peroxide compounds, making them effective in breaking down stubborn pollutants [12], [13]. A range of AOPs, such as Fenton, photo-Fenton, sonolysis, UV/O₃, and UV/H₂O₂, have been extensively studied [14], [15], [16], [17]. Nevertheless,

conventional Fenton-like processes come with inherent challenges. They are dosage-dependent, demanding large amounts of transition metals and oxidants to produce enough hydroxy radicals for effective treatment. Additionally, these processes require acidic conditions to prevent the formation of undesirable iron sludge precipitates [18], [19]. Even without sludge, removing excess dissolved iron and oxidants from treated water remains problematic, necessitating energy-intensive post-treatment.

In light of these challenges, catalytic wet peroxide oxidation (CWPO) using water-insoluble solid catalysts has emerged as a promising solution [16], [17], [20]. The hope is that this heterogeneous Fenton process can maintain high efficiency in decomposing organic pollutants over a broader pH range while allowing for easy recovery of the catalyst without sludge formation. Furthermore, one may expect that the CWPO reaction with *in-situ* generated H₂O₂ instead of read-made H₂O₂ may make it possible to realize an efficient wastewater treatment method enabling high degradation activity with minimized byproduct formation.

In this context, researchers have explored various approaches, such as photo-Fenton degradation of phenol in which the CdS/rGO composite was responsible for photocatalytic generation of H₂O₂, and subsequently added Fe²⁺ ion promoted the Fenton reaction [21]. However, these methods involved separate reactions and soluble iron salts. Similarly, electro-Fenton degradation of Rhodamine B using electro-generated H₂ and O₂ in the presence of a Pd/C catalyst was reported, albeit with homogeneous Fenton reaction using FeSO₄ [22]. Another electro-Fenton approach using heterogeneous Fe-ZSM-5 catalysts was also proposed [23]. The approach was involved with two steps such as electric H₂O₂ generation and Fenton reaction. Therefore, the second reaction should be initiated after electro-generated H₂O₂ concentration reached a certain level under an acidic condition. Additionally, Fenton-like oxidation of phenol using a Pd/Fe-ZSM-5 catalyst with formic acid as a hydrogen source and oxygen was explored [24]. The strategy was composed of three consecutive reactions; (i) formic acid dehydrogenation, (ii) direct synthesis of H₂O₂ from H₂ and O₂, and (iii) Fenton reaction with the *in-situ* generated H₂O₂. Despite these innovative approaches involving *in-situ* H₂O₂ generation, the use of

true heterogeneous catalysts for wastewater treatment in a concerted reaction system has been rarely realized.

In this study, we present a highly efficient and environmentally friendly catalytic wet peroxide oxidation (CWPO) of methyl orange (MO) using *in-situ* generated H₂O₂ from hydrogen and air instead of ready-made H₂O₂. To enhance *in-situ* H₂O₂ generation, we constructed a highly efficient Pt@Pd bimetallic catalyst on silica, involving the serial adsorption of different cationic and anionic metal precursors to form a sandwich-like metal complex salt, followed by Fe impregnation. As we will discuss, this unique bifunctional Fe-Pt@Pd/SiO₂ catalyst demonstrated outstanding activity in promoting CWPO with *in-situ* generated H₂O₂, even outperforming its counterpart with ready-made H₂O₂. These results clearly indicate that this innovative approach could be one of the most promising solutions for efficient water remediation, free from the addition and generation of undesirable chemicals that could affect the environment.

2. Experimental

2.1. Chemicals

Methyl Orange (C₁₄H₁₄N₃NaO₃S, dye content 85%), silica gel (SiO₂, 99%, 1.15 cm³/g pore volume), tetraamineplatinum(II) nitrate (Pt(NH₃)₄(NO₃)₂, 99%), potassium tetrachloropalladate(II) (K₂PdCl₄, 99%), potassium tetrachloroplatinate(II) (K₂PtCl₄, 98%), iron (II) sulfate heptahydrate (FeSO₄·7 H₂O, 99%), tetraoctylammonium bromide ([CH₃(CH₂)₇]₄N(Br), 98%), and sodium borohydride (NaBH₄, 98%) were supplied by Sigma-Aldrich (St. Louis, MO, USA). Tetraamineplatinum(II) nitrate solution (Pd(NH₃)₄(NO₃)₂, 5 wt% Pd) was obtained from Strem Chemicals (Newburyport, MA, USA). Hydrogen peroxide (H₂O₂, 34.5 wt%), methanol (CH₃OH, 99.8%), and sodium hydroxide (NaOH, 98%) were purchased from Samchun Chemicals (Seoul, South Korea). Dichloromethane (CH₂Cl₂, 99.5%) and cerium(IV) sulfate (Ce(SO₄)₂, 0.1 N standard solution) were supplied by Junsei Chemical (Tokyo, Japan) and Merck Millipore (Burlington, MA, USA), respectively. All chemicals were used as received without further purification.

2.2. Catalyst preparation

The Pt@Pd/SiO₂ catalyst was prepared through the sequential adsorption of two different cationic and anionic metal precursors on silica, forming a sandwich-like metal complex salt [25]. Initially, SiO₂ powder (0.5 g) was treated with 0.01 M NaOH (30 mL) for 30 min. It was then centrifuged and washed several times with deionized water until the pH of the supernatant reached 9.5–10. Subsequently, a Pd(NH₃)₄(NO₃)₂ aqueous solution (10 mL) was slowly added to the resulting SiO₂ dispersed water (20 mL) at a rate of approximately 0.05 mmol Pd/g-SiO₂. The mixed solution was stirred for 1 h, leading to the selective adsorption of the cationic Pd precursor onto the negatively charged SiO₂ surface. The resulting solid was isolated via centrifugation, washed twice with deionized water, and then dried at 333 K for 24 h. There was neither a color change nor precipitation in the remnant solution after centrifugation by the addition of NaBH₄ indicating the complete adsorption of the Pd precursor onto the silica.

The selective adsorption of the anionic Pt precursor onto the previously adsorbed cationic Pd precursor was carried out using a phase transfer technique. In brief, tetraoctylammonium bromide (TOABr, 0.037 mmol) and K₂PtCl₄ (0.017 mmol) were mixed in water (10 mL), followed by the addition of CH₂Cl₂ (20 mL). The resulting immiscible biphasic solution was vortexed for 5 min using a vortex mixer (SVM-10, Korea). After centrifugation, the water phase became discolored and was removed from the CH₂Cl₂ solution. The cationic Pd precursor-adsorbed SiO₂ was then added to the CH₂Cl₂ solution containing the phase-transferred anionic Pt precursor (Pd/Pt mole ratio = 1.1), and the mixture was stirred for 1 h. The resulting solid product was separated by centrifugation, washed with CH₂Cl₂, and dried at room temperature for 12 h. Subsequent calcination with air (100 mL/min) at 523 K for 3 h and reduction with 10 vol% H₂/N₂ (100 mL/min) at 473 K for 3 h led to the formation of the Pt@Pd/SiO₂ catalyst. It is noted that the expression 'Pt@Pd/SiO₂' does not mean any core-shell structure but simply indicates the order of metal precursors adsorbed onto the silica. Pd@Pt/SiO₂ catalyst was also prepared in a similar manner except for using Pt(NH₃)₄(NO₃)₂ and K₂PdCl₄ as cationic and anionic precursors, respectively.

For comparison, a range of monometallic or bimetallic catalysts was also prepared using the conventional incipient wetness impregnation (IWI) method with two cationic precursors, Pd(NH₃)₄(NO₃)₂ and/or Pt(NH₃)₄(NO₃)₂. For Pd/SiO₂ or Pt/SiO₂, a predetermined Pd or Pt precursor solution (metal content = 1.5 wt%) was slowly dropped onto SiO₂ with thorough mixing. After aging at 298 K for 12 h and drying at 393 K for 12 h, the Pd or Pt-impregnated silica was subjected to calcination and reduction, resulting in the formation of the Pd/SiO₂ or Pt/SiO₂ catalyst. In the case of PdPt/SiO₂, a mixed Pd and Pt precursor solution (total metal content = 1.5 wt% and Pd/Pt mole ratio = 1.1) was used. Fe was also impregnated onto naked SiO₂ and M/SiO₂ (M = Pd, Pt, PdPt, Pd@Pt, and Pt@Pd) catalysts via the IWI method using FeSO₄·7 H₂O (Fe intake 1 wt%) [26]. Afterward, calcination was carried out with air (100 mL/min) at 523 K for 3 h, followed by reduction using 10 vol% H₂/N₂ (100 mL/min) at 473 K for 3 h. The resulting catalysts were designated as Fe-M/SiO₂, i.e., Fe/SiO₂, Fe-Pd/SiO₂, Fe-Pt/SiO₂, Fe-PdPt/SiO₂, Fe-Pd@Pt/SiO₂, and Fe-Pt@Pd/SiO₂.

2.3. Characterization

UV-Visible diffuse reflectance spectra (UV-Vis DRS) were acquired using a Perkin-Elmer Lambda 25 spectrophotometer (Waltham, MA, USA) over a wavelength range of 300–800 nm at room temperature. Powder X-ray diffraction (PXRD) patterns were collected with a Malvern Panalytical Empyrean instrument (Malvern, UK). A Titan G2 ChemiSTEM (FEI company, USA) with an acceleration voltage of 200 kV was used to collect transmission electron microscopy (TEM) images, high-resolution TEM (HRTEM) images, and X-ray energy dispersive spectroscopy (EDS) elemental mapping measurements. Diffuse reflectance infrared Fourier transform spectroscopy (DRIFTS) was recorded using a Nicolet iS50 spectrometer (Thermo Fischer Science, USA) equipped with a mercury-cadmium-telluride A (MCT-A) detector and a diffuse reflectance chamber (Pike Technology, USA). Before carbon monoxide (CO) adsorption, a sample was reduced with 10 vol% H₂/N₂ (100 mL/min) at 473 K for 30 min. After sweeping the residual H₂ by purging with N₂ at an ambient temperature, 5 vol% CO/Ar gas was introduced into the DRIFTS cell for 15 min. DRIFTS data were collected at a resolution of 8 cm⁻¹ with 64 scans. X-ray

photoelectron spectroscopy (XPS) analyses were recorded on a Thermo VG Scientific (Waltham, MA USA) MultiLab 2000 with monochromatic Al K α at a pass energy of 50 eV. Binding energies were calibrated with a C 1 s peak at 284.6 eV.

2.4. Catalytic wet peroxide oxidation

The CWPO reaction of MO with *in-situ* generated H₂O₂ was carried out in the liquid phase under atmospheric pressure via two different ways, namely, CWPO-I using an M/SiO₂ catalyst with soluble Fe salt or CWPO-II using a bifunctional Fe-M/SiO₂ catalyst. In a typical CWPO-I run, 0.1 g of M/SiO₂ catalyst and 150 mL of MO (20 ppm) were introduced into a double-jacketed round-bottom flask (250 mL) connected to a circulator (WCR-P8, Daihan Scientific, Korea). After the temperature of the solution was stabilized at 293 K and residual gases were removed, 0.2 mM of FeSO₄·7 H₂O was added. The reaction was initiated by magnetic stirring at 600 rpm and maintained for 3 h. During the reaction, diluted hydrogen (5 vol% H₂/N₂) and model air (20 vol% O₂/N₂) were bubbled into the reaction solution (H₂/O₂/N₂ (vol%) = 3.8/5.0/91.2, 120 mL/min) by using two mass flow controllers (Brooks 5850E, Netherlands). CWPO-II reactions were performed in the same manner with a bifunctional Fe-M/SiO₂ catalyst in the absence of Fe salt. For comparison, catalytic activities of M/SiO₂ and Fe-M/SiO₂ catalysts for CWPO reactions in the presence of ready-made H₂O₂ were also evaluated. For instance, CWPO-I and CWPO-II reactions were performed after the addition of 5 mM ready-made H₂O₂ instead of hydrogen/air supply.

During the reaction, liquid samples were periodically taken from the reaction solution, filtered with a 0.45 μ m syringe filter to remove the catalyst particles. The MO concentration of an extracted sample was determined based on the intensity of the MO absorption peak at 464 nm using a UV-Visible spectrophotometer (Perkin-Elmer Lambda 25). For this purpose, a calibration curve was established with a series of known MO concentrations prior to the experiment ($R^2 = 0.999$). The MO conversion (%) and

degradation rate (mmol/h) were determined using (1), (2), respectively.

$$\text{MO Conversion}(\%) = \frac{\Delta C \times 100}{C_0} \quad (1)$$

$$\text{MO Degradation Rate}(\text{mmol} / \text{h}) = \frac{\Delta C \times V}{M_{\text{MO}} \times t} \quad (2)$$

where ΔC represents the difference between the initial MO concentration (C_0 , ppm) and that at a specific time (C_t , ppm), and V (L), M_{MO} , and t (h) are the total liquid volume (150 mL), MO molecular weight, and reaction time, respectively. The total organic carbon (TOC) content of the reaction product was analyzed by using TOC-V CPH TOC analyzer (Shimadzu Corp., Japan).

2.5. Direct synthesis, decomposition, and hydrogenation of H₂O₂

To examine the efficiency of a catalyst for *in-situ* generation of H₂O₂, direct synthesis of H₂O₂ (DSHP) from hydrogen and air was conducted in the presence of an M/SiO₂ catalyst under high pressure or atmospheric pressure conditions as described elsewhere [26]. In brief, M/SiO₂ catalyst (20 mg) and 80 wt% methanol/water mixed solution (50 mL) were poured into a jacketed high-pressure reactor (100 mL). Afterward, the reactor was pressurized with two mixed gases (5 vol% H₂/N₂ and 20 vol% O₂/N₂) up to 30 bar at 275 K. Once the reactor pressure and temperature were stabilized, a reaction was initiated by vigorous stirring and maintained for 1 h. During the reaction, the mixed gases were continuously supplied to the reactor (total flow rate = 40 mL/min), and the mole ratio of H₂/O₂/N₂ was fixed the same as that for the CWPO reaction. The H₂O₂ produced was titrated against acidic cerium (IV) sulfate solution with ferroin indicator. To determine the changes in H₂ concentration during the reaction, off-gas was collected in a gas bag and analyzed with a gas chromatograph (Agilent 6890, USA) equipped with a Carboxen-1010 PLOT capillary column and a thermal conductivity detector.

The degree of H₂O₂ decomposition and hydrogenation promoted by a catalyst was evaluated at 275 K under atmospheric pressure. For the H₂O₂ decomposition test, a 1 wt% H₂O₂ methanol/water solution (50 g) was stirred in the presence of a catalyst (20 mg) at

275 K for 1 h under an inert atmosphere. H₂O₂ hydrogenation was performed in a similar manner except for H₂ bubbling (5 mL/min) instead of inert gas purging. During the H₂O₂ decomposition and hydrogenation tests, reaction solutions were periodically sampled and titrated to examine the changes in H₂O₂ concentration.

It is noted that 80 wt% methanol/water solution was used for the H₂O₂ formation (DSHP), decomposition, and hydrogenation because the use of mixed methanol/water solvent is better to compare the H₂O₂ productivity of each catalyst, especially under atmospheric pressure, due to the high solubility of H₂ in methanol. In this context, the reaction temperatures of the DSHP and H₂O₂ decomposition/hydrogenation tests were also lowered to 275 K because H₂O₂ productivity can increase at a lower temperature, and therefore the lower temperature is beneficial in examining the difference in the reaction performance of each catalyst.

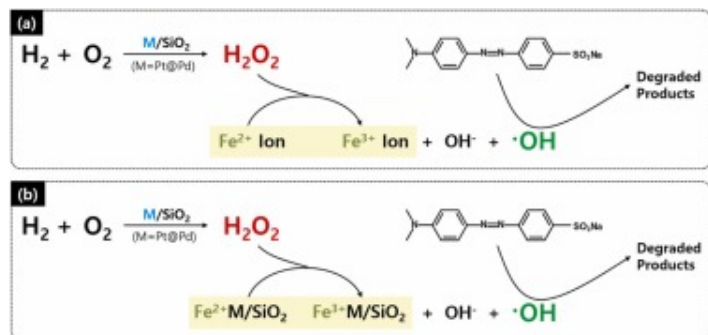
3. Results and discussion

3.1. Strategy for CWPO with *in-situ* generated H₂O₂

Although the combination of CWPO reaction with *in-situ* generation of H₂O₂ is fascinating from environmental and economical viewpoints, the one-pot reaction is hardly achievable through most photo-Fenton and electro-Fenton reactions. Although H₂O₂ can be generated *in-situ* in the same reactor from hydrogen and oxygen sources, H₂O₂ should be pre-generated by O₂ bubbling to reach a certain level in the presence of a photocatalyst or electrodes. Furthermore, an iron catalyst such as ferrous sulfate (FeSO₄) can be added after stopping the O₂ bubbling to avoid Fe²⁺ being oxidized to Fe³⁺ [21], [23]. Because the subsequent Fenton reaction proceeds with the 'ready-made' H₂O₂, the approach becomes similar as the conventional Fenton reaction, making it difficult to maintain the reaction after depletion of the ready-made H₂O₂.

In this regard, to realize an efficient and recyclable CWPO reaction without the addition and generation of any chemicals, it is aimed here to combine the Fe-promoted CWPO

reaction with the noble metal-catalyzed direct synthesis of H₂O₂ (DSHP) from hydrogen and air for *in-situ* generation of H₂O₂ in a concerted manner. To evaluate the effect of these two different catalysts on the specific and overall reaction rates, two CWPO approaches were adopted. As described in Fig. 1, the CWPO-I reaction was performed in the presence of M/SiO₂ and soluble Fe salt (FeSO₄). On the other hand, the CWPO-II reaction was conducted with a bifunctional Fe-M/SiO₂ catalyst prepared by the impregnation of Fe onto the M/SiO₂.



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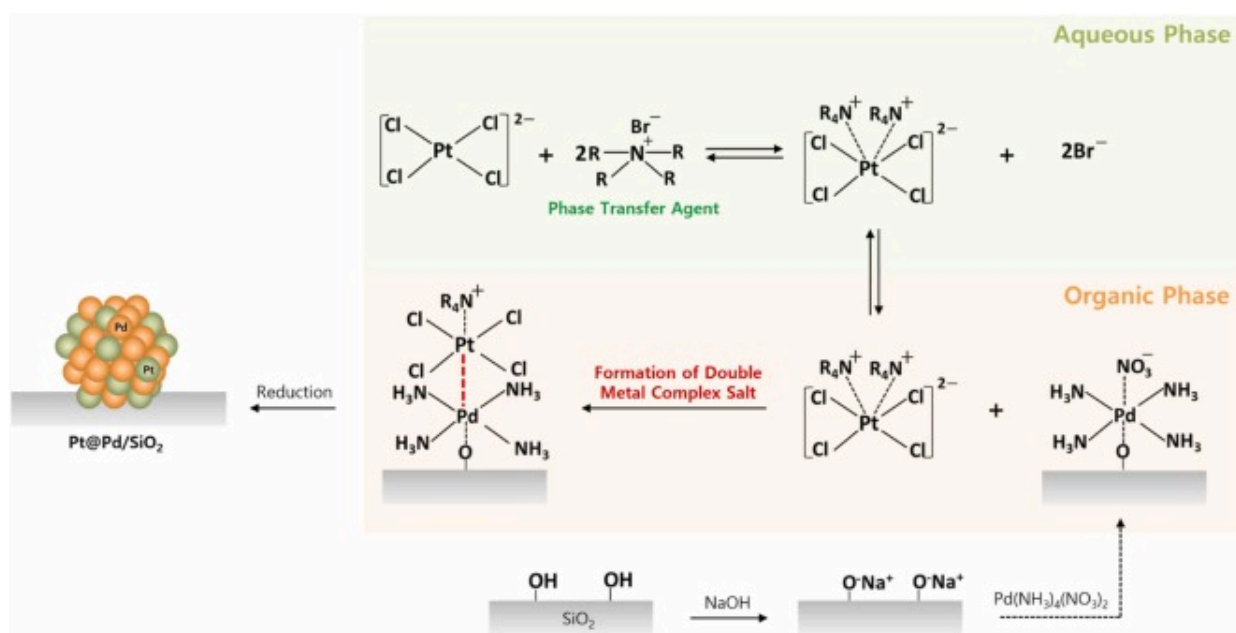
Fig. 1. CWPO of MO with *in-situ* generated H₂O₂: (a) CWPO-I with a M/SiO₂ (M = Pd, Pt, PdPt, Pd@Pt, or Pt@Pd) catalyst and Fe salt; (b) CWPO-II with a Fe-M/SiO₂ catalyst.

3.2. Catalyst preparation and characterization

Because the degradation rate of MO is highly dependent on the H₂O₂ production rate, designing an efficient catalyst for the DSHP reaction is crucial. Therefore, our first study was focused on preparing a supported noble metal catalyst, which can efficiently facilitate the H₂O₂ production. For this purpose, silica supported bimetallic Pt@Pd and Pd@Pt catalysts were prepared by double-metal complex salt approach. Because the Pt and Pd ions can exist very closely each other in such sandwich-like metal complex, it was assumed that the approach would be more advantageous for the PtPd alloy formation compared to the conventional co-adsorption method where competitive adsorption of

different metal ions and the formation of a certain portion of single metal nanoparticles (NP) are inevitable [25], [27].

As shown in Fig. 2, the preparation of Pt@Pd/SiO₂ catalyst comprises four steps: (i) ion exchange of surface silanol groups of silica, (ii) selective adsorption of a cationic Pd precursor ($[\text{Pd}(\text{NH}_3)_4]^{2+}$) on the negatively charged silica surface, (iii) formation of Pt-Pd heterometallic complex salt by the selective adsorption of phase-transferred anionic Pt precursor ($(\text{R}_4\text{N}^+)_2[\text{PtCl}_4]^{2-}$, where R_4N^+ = tetraoctylammonium) on the pre-adsorbed cationic Pd precursor, and (iv) heat treatments (calcination/reduction) for the formation of bimetallic NPs. The pH of the slurry was important in the second step, and efficient and quantitative adsorption of the cationic Pd precursor ($[\text{Pd}(\text{NH}_3)_4]^{2+}$) on the negatively charged silica surface was possible between pH 9.5–10. Above pH 10, the quantitative adsorption of the Pd precursor was less reproducible owing to its high ionic strength under the highly basic condition [28]. It is noted that a more sophisticated approach is needed for the construction of the Pt-Pd heterometallic complex on silica. Considering that most of heterometallic complex salts are insoluble and hardly formed in water [29], [30], direct adsorption of the anionic Pt precursor ($[\text{PtCl}_4]^{2-}$) on the $(\text{NO}_3^-)[\text{Pd}(\text{NH}_3)_4]^{2+}\text{O}^-/\text{SiO}_2$ in aqueous solution resulted in the formation of large aggregates and/or monometallic NPs [25]. To circumvent the problem, the anionic Pt precursor was transferred from aqueous phase to organic phase (CH_2Cl_2) by means of phase-transfer technique. As shown in Fig. S1, both Pt and Pd precursors could be quantitatively transferred as $(\text{R}_4\text{N}^+)_2[\text{PtCl}_4]^{2-}$ and $(\text{R}_4\text{N}^+)_2[\text{PdCl}_4]^{2-}$, respectively. The adsorption of the phase-transferred $(\text{R}_4\text{N}^+)_2[\text{PtCl}_4]^{2-}$ on the $(\text{NO}_3^-)[\text{Pd}(\text{NH}_3)_4]^{2+}\text{O}^-/\text{SiO}_2$ in the organic solvent successfully formed Pt-Pd heterometallic complex salt on silica $((\text{R}_4\text{N}^+)[\text{PtCl}_4]^{2-}[\text{Pd}(\text{NH}_3)_4]^{2+}\text{O}^-/\text{SiO}_2)$ by virtue of the metallophilic bonding between the electrophile of metallic cation and the nucleophile of metallic anion.



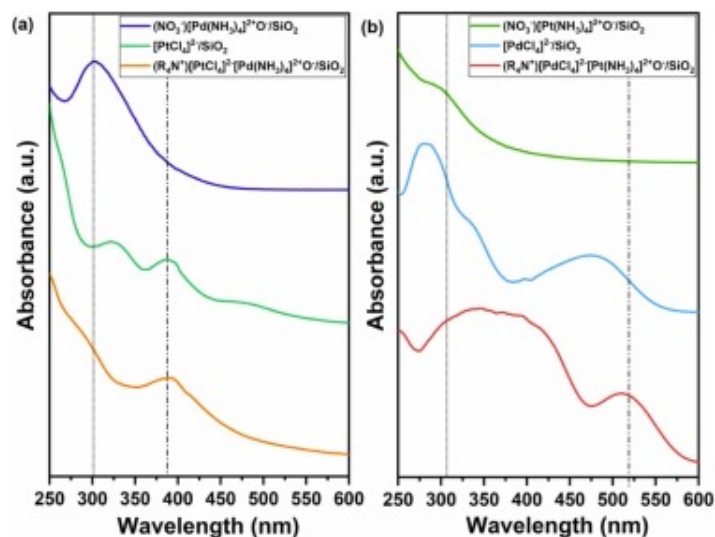
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Fig. 2. Preparation of bimetallic Pt@Pd/SiO₂ catalyst via double-metal complex salt approach.

The sequential adsorption of two different cationic and anionic metal precursors on silica was examined by UV-Vis DRS technique, and the results are presented in Fig. 3. The weak absorption peaks at around 300–330 nm are ascribed to the presence of square planar Pd- and Pt-amine complexes on silica through d-d transitions [25] as well as associated with a transition of NO₃⁻ indicating the formation of (NO₃⁻)[Pd(NH₃)₄]²⁺O⁻/SiO₂ and (NO₃⁻)[Pt(NH₃)₄]²⁺O⁻/SiO₂ [31]. On the other hand, the strong absorption peaks observed below 300 nm for [PtCl₄]²⁻/SiO₂ and [PdCl₄]²⁻/SiO₂ may be attributed to a ligand-to-metal charge transfer [32]. In the cases of (R₄N⁺)[PtCl₄]²⁻[Pd(NH₃)₄]²⁺O⁻/SiO₂ (Fig. 3(a)) and (R₄N⁺)[PdCl₄]²⁻[Pt(NH₃)₄]²⁺O⁻/SiO₂ (Fig. 3(b)), there exist two broad absorption band at ca. 390 and 520 nm, respectively, which are attributed to the formation of Pt-Pd heterometallic complex salt on silica [25]. The formation of bimetallic Pt-Pd complex salt can also be

confirmed by the color change of the solid. For example, while there was no appreciable color for $(\text{NO}_3^-)[\text{Pd}(\text{NH}_3)_4]^{2+}\text{O}^-/\text{SiO}_2$, it turned light pink after the adsorption of $(\text{R}_4\text{N}^+)_2[\text{PtCl}_4]^{2-}$ [25], [33]. The result implies the formation of $(\text{R}_4\text{N}^+)[\text{PtCl}_4]^{2-}[\text{Pd}(\text{NH}_3)_4]^{2+}\text{O}^-/\text{SiO}_2$ likewise Magnus' green salt ($[\text{Pt}(\text{NH}_3)_4]^{2+}[\text{PtCl}_4]^{2-}$) or Vauquelin's pink salt ($[\text{Pd}(\text{NH}_3)_4]^{2+}[\text{PdCl}_4]^{2-}$) [29].



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Fig. 3. UV-Vis DRS of Pt-Pd heterometallic complex salts in the order of their construction on silica: (a) $(\text{R}_4\text{N}^+)[\text{PtCl}_4]^{2-}[\text{Pd}(\text{NH}_3)_4]^{2+}\text{O}^-/\text{SiO}_2$ and (b) $(\text{R}_4\text{N}^+)[\text{PdCl}_4]^{2-}[\text{Pt}(\text{NH}_3)_4]^{2+}\text{O}^-/\text{SiO}_2$.

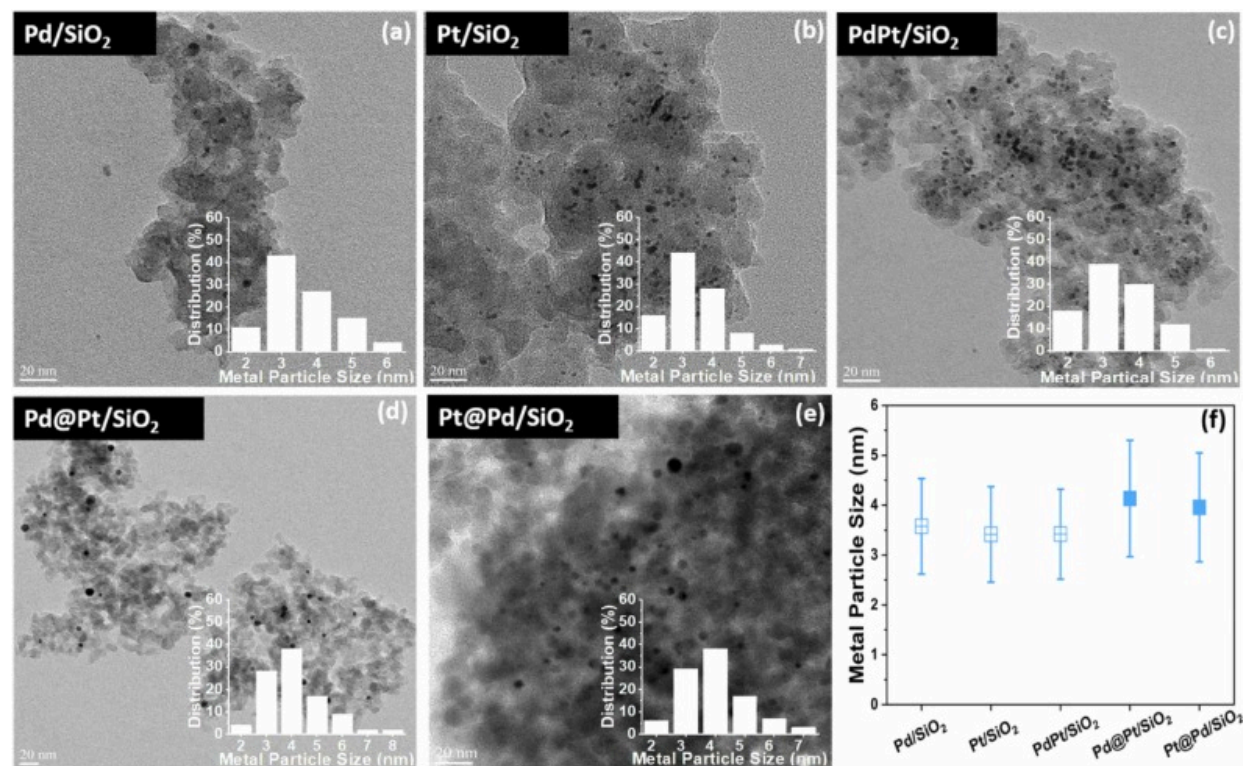
The existence of quatarnary ammonium cations in the double complex salts was also examined by comparing the N 1s XPS profile of the cationic $[\text{Pd}(\text{NH}_3)_4]^{2+}$ complex adsorbed silica with those of silica supported Pt-Pd heterometallic complex salts by altering the adsorption order of two different metal precursors. As shown in Fig. S2, the appearance of two distinct peaks centered approximately at 400 and 402.5 eV for both $(\text{R}_4\text{N}^+)[\text{PtCl}_4]^{2-}[\text{Pd}(\text{NH}_3)_4]^{2+}\text{O}^-/\text{SiO}_2$ and $(\text{R}_4\text{N}^+)[\text{PdCl}_4]^{2-}[\text{Pt}(\text{NH}_3)_4]^{2+}\text{O}^-/\text{SiO}_2$ correspond to the N atoms of amine and quatarnary ammonium cations, respectively [34], [35].

However, in the case of (NO₃⁻)[Pd(NH₃)₄]²⁺O⁻/SiO₂, only the N 1 s peak contributed from -NH₃ group is observed.

The crystallinity of the Pt@Pd and Pd@Pt NPs after a series of heat treatment of the resulting Pt-Pd heterometallic complex salts was examined by PXRD analysis. [Fig. S3](#) presents the PXRD patterns of various silica supported monometallic and bimetallic noble metal NPs prepared either by the double metal complex approach or the conventional IWI method. For all catalysts, typical diffraction peaks of metallic Pt and Pd, i.e., (111), (200), and (220) planes, originating from a face-centered cubic (fcc) lattice were observed [\[36\]](#), [\[37\]](#). The lattice constant parameter and constraint of bimetallic NPs were also calculated using the Debye-Scherrer equation [\[38\]](#). As summarized in [Table S1](#), the Bragg's angles (2θ) observed for the Pt@Pd/SiO₂, Pd@Pt/SiO₂, and PdPt/SiO₂ catalysts were found to lie between the (111) peaks of monometallic Pt and Pd at 39.36° and 40.01°, respectively. The lattice constraint of bimetallic NPs highly depended on the preparation method in the order of Pt@Pd/SiO₂ (1.418%) > Pd@Pt/SiO₂ (1.371%) >> PdPt/SiO₂ (0.195%). Since the changes in the lattice constraint are closely related to the degree of alloy formation [\[39\]](#), [\[40\]](#), it seems that the construction of the Pt-Pd heterometallic complex salt on silica was superior to the conventional co-adsorption of two different metal precursors in terms of more efficient Pt-Pd alloy formation.

[Fig. 4](#) shows the TEM images of M/SiO₂ catalysts with their NP size distribution. The average NP sizes of the Pt@Pd/SiO₂ and Pd@Pt/SiO₂ were slightly larger than those of others prepared by the conventional IWI method. However, there was little aggregation for Pt@Pd/SiO₂ and Pd@Pt/SiO₂ compared to PdPt/SiO₂. The good metal dispersion achieved by the double metal complex salt approach may be ascribed to the beneficial role of quaternary ammonium ion (R₄N⁺). In addition to its crucial role as a phase-transfer agent, the R₄N⁺ ion can act as a capping agent and efficiently suppress an undesirable sintering or aggregation of metal NPs during a heat treatment process [\[41\]](#), [\[42\]](#). HR-TEM analysis was also conducted to examine the lattice spacing of the monometallic and bimetallic NPs. As provided in [Fig. S4](#), the interplanar spacing of Pd/SiO₂ and Pt/SiO₂ was 0.2238 and 0.2380 nm, respectively, corresponding to the (111) plane of fcc-structured Pd

and Pt, respectively. In the cases of the bimetallic NPs, their lattice spacing values were between those of each monometallic counterpart, indicating the formation of PtPd bimetallic alloys regardless of the preparation method [43], [44].



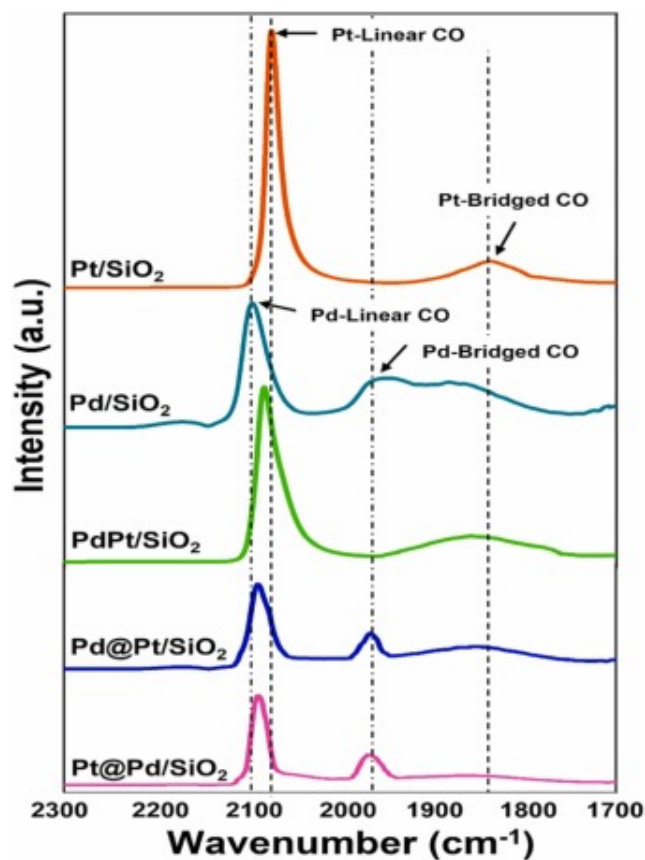
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Fig. 4. (a-e) TEM images of various M/SiO₂ catalysts prepared by either double metal complex approach or conventional IWI method and (f) NP size distributions of catalysts.

Given that the IR absorption band position of CO is highly sensitive to the properties of adsorption sites, such as the surface of nanoparticles (NPs) [45], [46], [47], [48], CO-DRIFTS study was conducted to investigate changes in the IR band position of adsorbed CO on silica-supported monometallic and bimetallic noble metal NPs. As shown in Fig. 5, the IR band position of CO depended not only on the type of metal NPs but also on the

preparation method. In the case of Pt/SiO₂, the intense peak at 2075 cm⁻¹ indicated the adsorption of linear CO on reduced Pt sites, while the weak and broad band at around 1836 cm⁻¹ suggested the presence of Pt-bridged CO species. CO-DRIFTS analysis of Pd/SiO₂ also revealed two distinct peaks at 2094 and 1962 cm⁻¹, indicating the existence of linear and bridged CO species on Pd, respectively.



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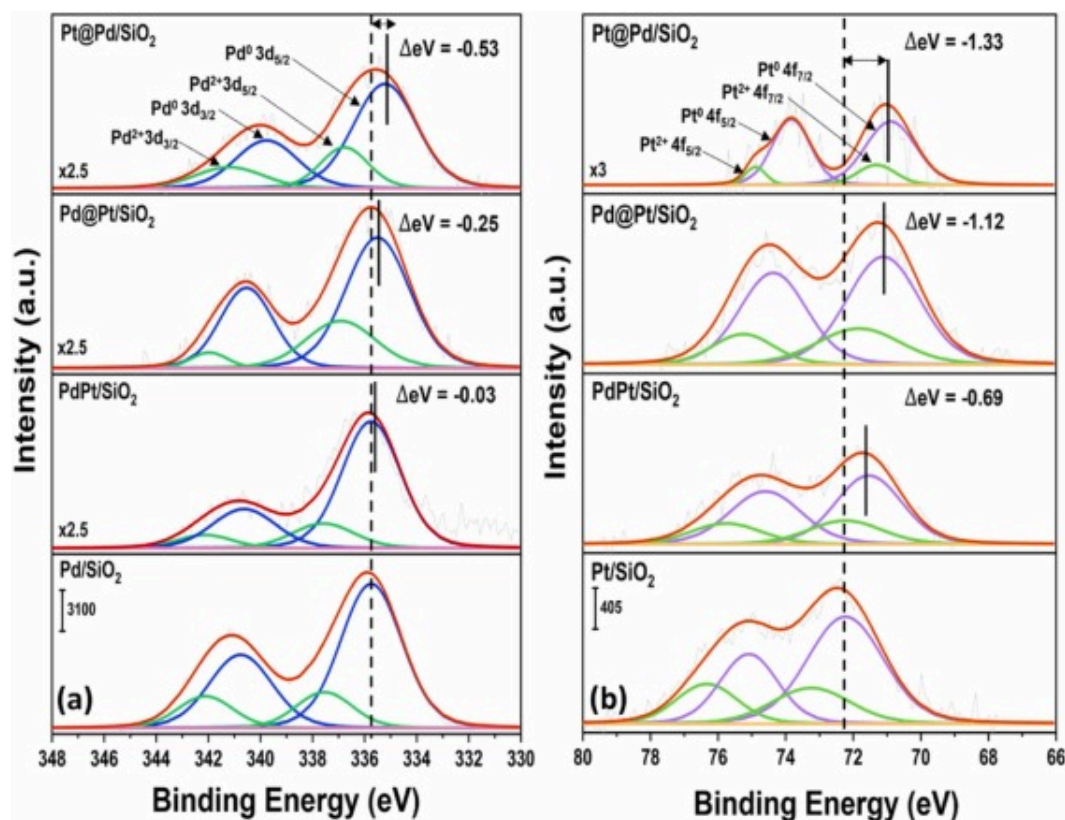
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Fig. 5. DRIFTS of adsorbed CO on various silica supported monometallic and bimetallic noble metal NPs.

For bimetallic NPs, the strong absorption band corresponding to linear CO adsorption on metal shifted to a position between those for CO-Pd and CO-Pt, confirming the formation of Pt-Pd bimetallic NPs. However, it is important to note that the CO-DRIFTS of PdPt/SiO₂ differed from those of Pt@Pd/SiO₂ and Pd@Pt/SiO₂. In the case of PdPt/SiO₂, the absorption band position of linear CO-M closely resembled that of Pt/SiO₂, and the broad IR band for Pt-bridged CO species was also observed. In contrast, for Pt@Pd/SiO₂ and Pd@Pt/SiO₂, the linear CO-M band shifted to a higher wavenumber, approaching that of CO-Pd. Additionally, the IR band for Pd-bridged CO species was clearly observed, with a weakening of the Pt-bridged one. These results clearly indicate that Pd was largely exposed on the surface of Pt@Pd and Pd@Pt bimetallic NPs prepared by the double complex salt approach. On the other hand, the conventional IWI method led to the formation of PtPd bimetallic NPs with a Pt-enriched surface.

To examine the electronic properties of Pd and Pt species within Pt@Pd and Pd@Pd bimetallic NPs, XPS analysis was performed, and the results are presented in [Fig. 6](#). For the Pd 3d XPS ([Fig. 6\(a\)](#)), two strong deconvoluted peaks at approximately 335 and 340 eV corresponded to the Pd 3d_{5/2} and Pd 3d_{3/2} states of Pd⁰, respectively [[49](#)], [[50](#)]. In addition, two weak deconvoluted peaks of Pd²⁺ were observed as a spin-orbit doublet at around 337.5 eV for 3d_{5/2} and 342 eV for 3d_{3/2}, respectively. These findings indicate the highly metallic nature of Pd species, regardless of the catalyst preparation method. Moreover, the shift of Pd 3d XPS peaks toward lower binding energy (BE) may be indicative of bimetal formation. As shown in [Fig. 6\(a\)](#), the BE of Pd 3d peaks shifted to more negative values in the order of PdPt/SiO₂ (- 0.03 eV) < Pd@Pt/SiO₂ (- 0.25 eV) < Pt@Pd/SiO₂ (- 0.53 eV) compared to that of Pd/SiO₂. The results indicate that the catalyst preparation method plays a significant role in modulating the electronic state of metal species on the silica support. The larger BE shift of the Pt@Pd and Pd@Pd bimetallic NPs implies a stronger interaction between Pd and Pt species than that within PdPt NPs. As shown in [Fig. 6\(b\)](#), the appearance of strong Pt 4f_{5/2} and Pt 4f_{7/2} peaks of Pt⁰, along with relatively weaker Pt 4f_{5/2} and Pt 4f_{7/2} peaks of Pt²⁺ also indicates a high content of metallic Pt within monometallic and bimetallic NPs. The BE shifts of Pt 3f XPS peaks were

also observed for the Pt@Pd, Pd@Pd, and PdPt bimetallic NPs in the same order as the Pd 3d XPS: PdPt/SiO₂ (-0.69 eV) < Pd@Pt/SiO₂ (-1.12 eV) < Pt@Pd/SiO₂ (-1.33 eV) compared to that of Pt/SiO₂. The larger BE shift of Pt 4f peaks, compared to those of Pd 3d, can be explained by electron transfer from Pd to Pt due to the higher electronegativity of Pt (2.28) than Pd (2.20) [44], [51].



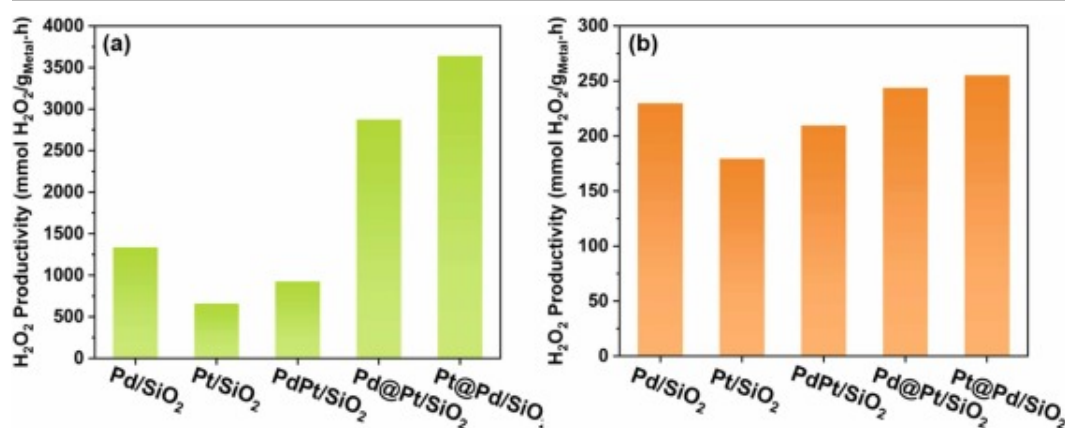
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Fig. 6. (a) Pd 3d and (b) Pt 4f XPS spectra of various M/SiO₂ catalysts.

3.3. Direct synthesis of H₂O₂ (DSHP)

The potential of the as-prepared catalysts for H₂O₂ production was evaluated by examining their reaction performance in the DSHP reaction. Since a high-pressure condition is beneficial in accelerating the H₂O₂ production rate due to the poor solubility of H₂ in a polar solvent [52], [53], and therefore in examining the difference in catalytic activity more clearly, the DSHP reaction was conducted at 30 bar in the absence of any caustic acid and/or halide additives. As depicted in Fig. 7(a), it became evident that not only the type of noble metal NPs but also the preparation method played a crucial role in enhancing H₂O₂ production. Pd is widely recognized as one of the most efficient metals for promoting the DSHP reaction [53], [54], and Pd/SiO₂ exhibited better activity than Pt/SiO₂. Additionally, the increased H₂O₂ productivity with PdPt/SiO₂ compared to Pt/SiO₂ might be attributed to the ensemble effect arising from PdPt alloy formation. However, the lower reaction performance of PdPt/SiO₂ compared to Pd/SiO₂ indicates that the synergistic effect due to bimetal formation was insufficient to surpass the activity of Pd/SiO₂. It appears that the Pt-enriched nature of the PdPt NP surface weakened the beneficial effect of bimetal formation.



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Fig. 7. DSHP reaction over various M/SiO₂ catalysts: (a) under a high pressure (30 bar) and (b) atmospheric pressure.

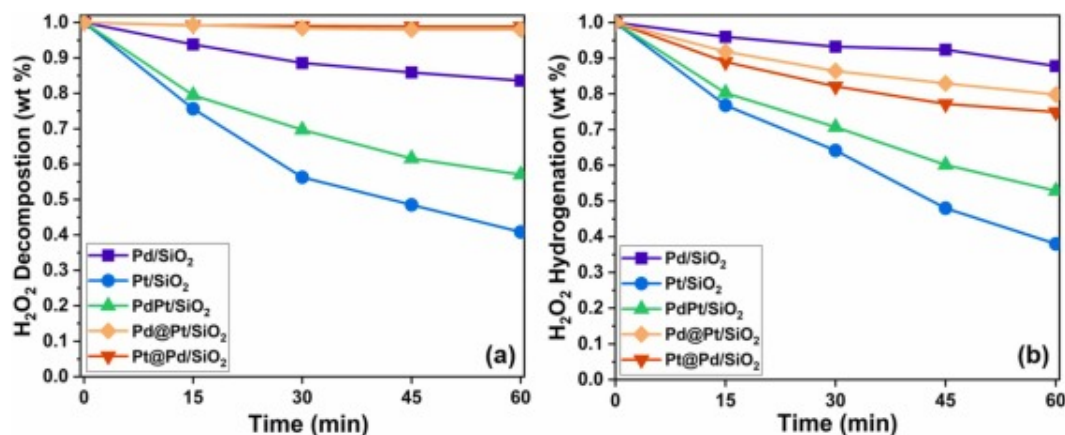
Conversely, the bimetallic catalysts prepared using Pd-Pt heterometallic complex salts exhibited superb activity in promoting the DSHP reaction. The H₂O₂ productivities of Pd@Pt/SiO₂ and Pt@Pd/SiO₂ were 3.1 and 3.9 times higher, respectively, than that of PdPt/SiO₂. The highest H₂O₂ productivity, 3642 mmol-H₂O₂/g-metal.h, obtained with the Pt@Pd/SiO₂ catalyst, is approximately three times higher even when compared to Pd/SiO₂. This superior H₂O₂ productivity can be explained by the strong interaction between Pt and Pd atoms within the Pt@Pd NP domain, and the highly Pd-enriched nature of the Pt@Pd NP surface, as confirmed by XPS and CO-DRIFTS analyses, respectively. These results clearly demonstrate that the catalyst preparation method had a significant impact on determining the characteristics of alloyed NPs. The double-metal complex method was superior in maximizing the synergistic effect during the formation of Pd-Pt bimetallic NPs, and consequently, in boosting H₂O₂ production.

In the case of the DSHP reaction under atmospheric pressure, as shown in Fig. 7(b), the H₂O₂ production rate was significantly reduced regardless of the catalysts used. This reduction occurred because the amount of dissolved hydrogen in solution would be very small under atmospheric pressure. While there was no change in the order of catalytic activity for H₂O₂ production, the difference in H₂O₂ productivity between catalysts was greatly reduced, indicating that the DSHP reaction was considerably impeded by the low hydrogen concentration in the reaction solution under atmospheric pressure.

One of the challenging problems in the DSHP reaction is that all the involved reaction routes are thermodynamically feasible, and the active metal may promote not only H₂O₂ production but also the sequential decomposition and hydrogenation of the ready-generated H₂O₂ as described in Fig. S5. Therefore, the overall H₂O₂ production can be attained as a result of reflecting the amount of generated, decomposed, and hydrogenated H₂O₂ from the DSHP and subsequent decomposition/hydrogenation on the same active sites.

In this context, the reaction performance of the various M/SiO₂ catalysts for H₂O₂ decomposition and hydrogenation was assessed. As revealed in Fig. 8, the degree of H₂O₂

decomposition and hydrogenation was largely affected by the features of the catalysts and matched well with the reaction performance of each catalyst for the DSHP reaction. For instance, the highest H₂O₂ decomposition and hydrogenation occurred in the presence of Pt/SiO₂ catalyst, followed by PdPt/SiO₂, suggesting the inefficiency of Pt NPs in suppressing unfavorable side-reactions. Pt or Pt-enriched PdPt NPs readily promote the unfavorable dissociation of oxygen in H-OOH and therefore facilitate H₂O₂ decomposition to H₂O [45].



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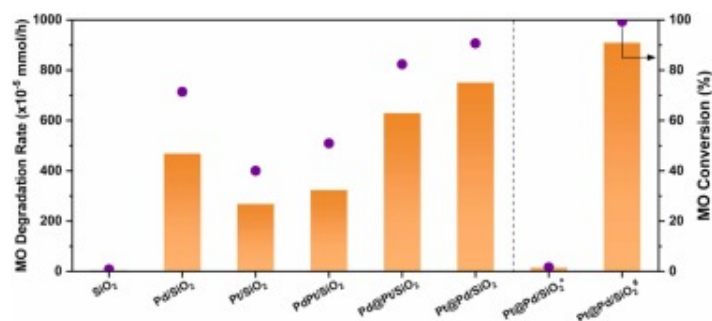
Fig. 8. (a) H₂O₂ decomposition and (b) H₂O₂ hydrogenation over various M/SiO₂ catalysts.

On the contrary, there was little H₂O₂ decomposition by Pd@Pt/SiO₂ or Pt@Pd/SiO₂ catalyst, which explains their excellent activity in H₂O₂ production (Fig. 8(a)). These catalysts were also efficient in suppressing H₂O₂ hydrogenation comparable to that of Pd/SiO₂ (Fig. 8(b)). Considering that the subsequent H₂O₂ decomposition and hydrogenation mainly determine the overall H₂O₂ productivity, and the use of caustic additives has been considered inevitable for the DSHP reaction to suppress undesirable side-reactions [52], [53], the greatly reduced activities of Pd@Pt/SiO₂ and Pt@Pd/SiO₂ catalysts for H₂O₂ decomposition and hydrogenation without the use of any corrosive

promoters are highly promising features in designing an efficient catalyst for *in-situ* production of H₂O₂.

3.4. CWPO-I

The CWPO-I reaction was conducted in the presence of an M/SiO₂ catalyst and soluble iron salt (FeSO₄). It is noted that a heterogeneous Fenton process using a water-insoluble solid catalyst is generally referred to as CWPO. Therefore, the definition of 'CWPO-I' used here closely resembles a Fenton or Fenton-like reaction due to the use of soluble iron salt. However, the 'CWPO-I' expression is used to distinguish this unique catalysis, which involves both a heterogeneous DSHP catalyst (M/SiO₂) and a homogeneous iron catalyst, from the conventional Fenton reaction. As presented in Fig. 9, the MO degradation rate is highly dependent on the catalyst used and follows the order of Pt/SiO₂ < PdPt/SiO₂ < Pd/SiO₂ < Pd@Pt/SiO₂ < Pt@Pd/SiO₂, which aligns perfectly with their H₂O₂ production trends. This result clearly demonstrates that two different reactions, DSHP and CWPO, can be successfully carried out consecutively. Furthermore, it highlights that the *in-situ* H₂O₂ generation step in DSHP predominantly influences MO degradation efficiency, as depicted in Fig. 10.

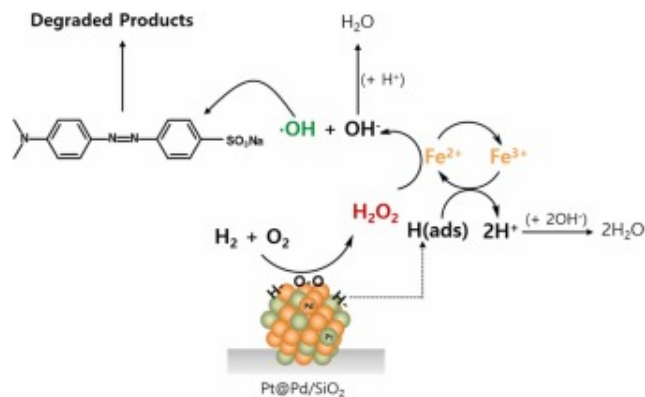


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Fig. 9. CWPO-I of MO with *in-situ* generated H₂O₂. Reaction condition: catalyst (0.1 g), MO (20 ppm), [Fe²⁺] (0.2 mM), H₂O (150 mL), H₂/O₂/N₂ (mol%) = 3.8/5.0/91.2 (total flow rate =

120 mL/min), 293 K, 1 bar, reaction time = 3 h; * without mixed gas nor H₂O₂; # ready-made H₂O₂ (5 mM) was used instead of mixed gas.



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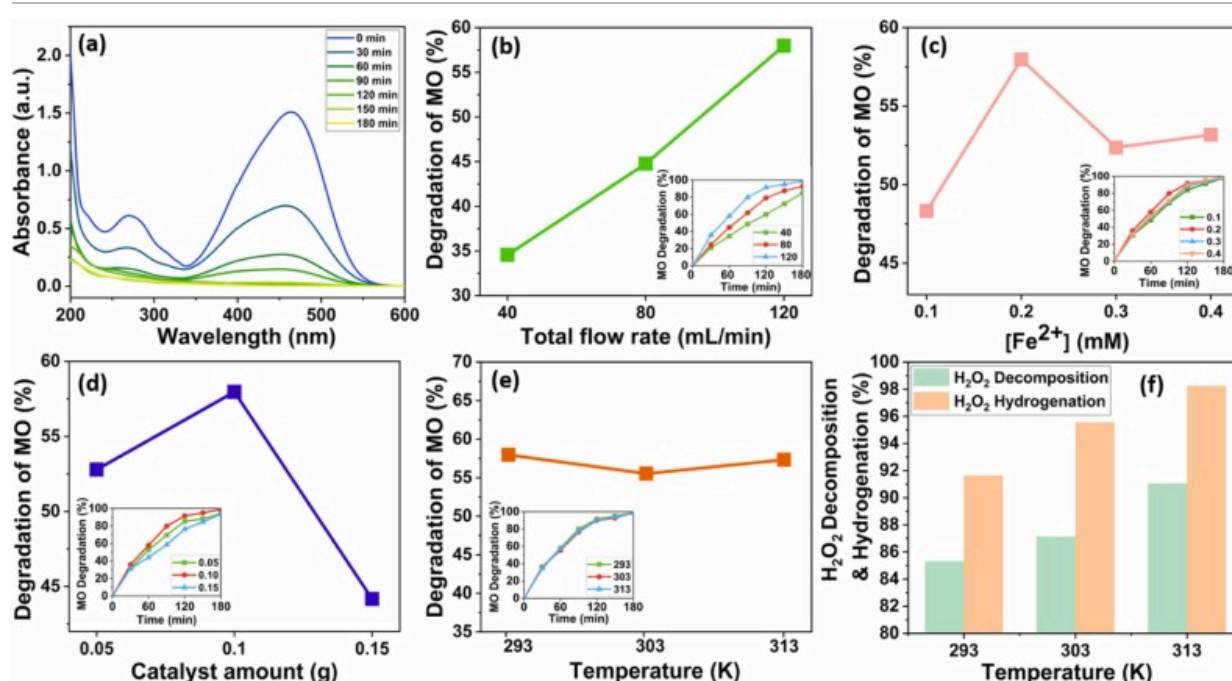
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Fig. 10. Postulated reaction mechanism of the CWPO-I consisting of the M/SiO₂-promoted DSHP and sequential homogenous Fe-triggered Fenton reaction with the *in-situ* generated H₂O₂.

However, the faster MO degradation rate achieved using ready-made H₂O₂ suggests that a high H₂O₂ concentration in the solution is beneficial for rapidly generating a sufficient amount of hydroxyl radicals through H₂O₂ decomposition. As shown in [Fig. S6](#), MO decomposed gradually with *in-situ* generated H₂O₂, but decomposed within 1 h when ready-made H₂O₂ (5 mM) was added. This result reaffirms that the Fenton reaction, facilitated by dissolved Fe²⁺ ions, is very fast, and the H₂O₂ productivity of a catalyst for the DSHP reaction primarily determines the overall MO degradation rate.

To examine the factors affecting the reaction performance of the Pt@Pd/SiO₂ catalyst for MO degradation, a parameter study was conducted for some key factors. [Fig. 11](#) (a) shows the UV-Vis absorption spectra of MO during the reaction. As the CWPO-I reaction progressed, the MO absorption band gradually decreased and disappeared after 3 h,

indicating complete MO degradation. No additional absorption bands except for the characteristic absorption band of MO were observed during the CWPO-I reaction. The effect of the total flow rate of H₂/O₂/N₂ mixed gas on MO degradation is provided in [Fig. 11 \(b\)](#). There is a clear linear relationship between MO degradation and the total flow rate, emphasizing the crucial role of H₂O₂ production rate in facilitating MO degradation. While increasing the iron concentration also enhanced MO degradation, there was little additional improvement above 0.3 mM of [Fe²⁺] ([Fig. 11 \(c\)](#)). Moreover, [Fig. 11 \(d\)](#) shows a volcano-shaped dependency between MO degradation and the amount of Pt@Pd/SiO₂. The reduced MO degradation at higher catalyst amounts may be due to decreased H₂O₂ productivity, possibly caused by enhanced H₂O₂ hydrogenation as the active metal content increased [55]. [Fig. 11\(e\)](#) clearly demonstrates that MO degradation was hardly enhanced by raising the temperature. This suggests that the reaction temperature has little influence on both the Fenton reaction and DSHP reaction. While higher temperatures can enhance the H₂O₂ formation reaction (DSHP), the sequential H₂O₂ decomposition and hydrogenation will be accelerated further, as evidenced in [Fig. 11\(f\)](#). The trade-off effect of the reaction temperature on H₂O₂ production accounts for the weak relationship between MO degradation and reaction temperature.



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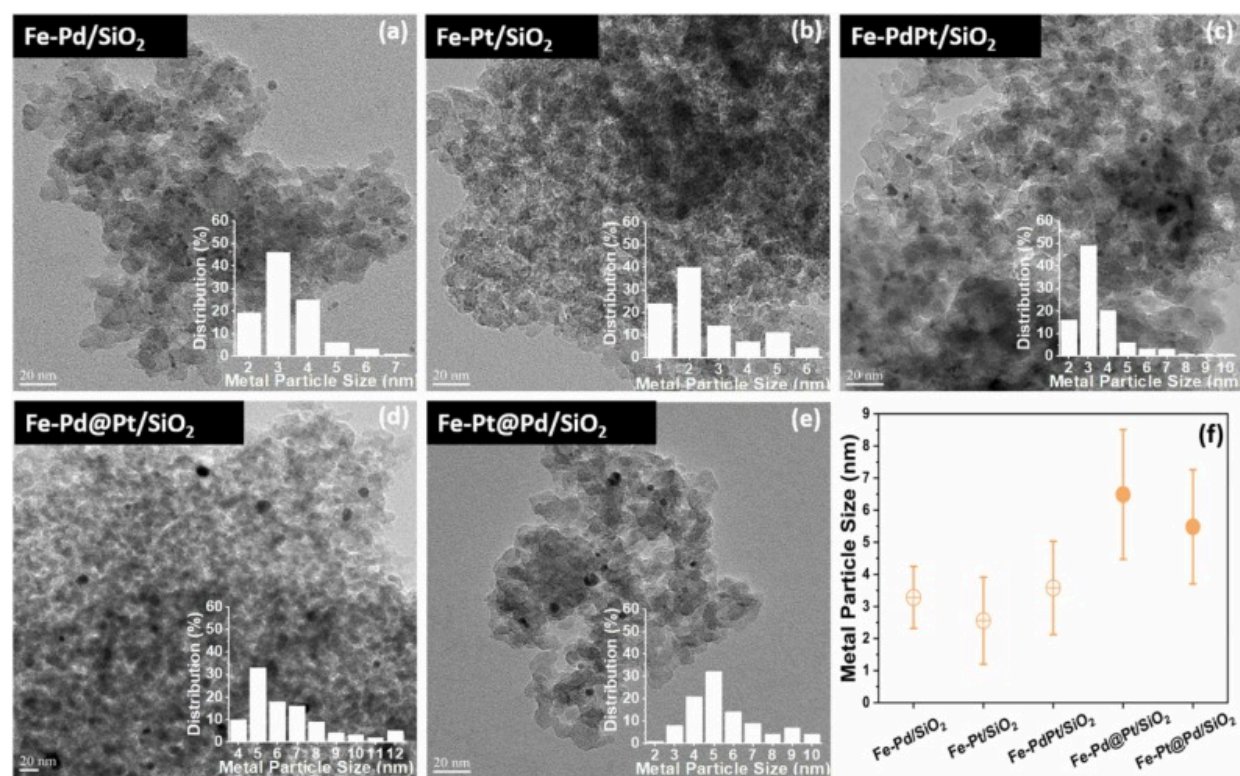
Fig. 11. Parameter study for the CWPO-I reaction in the presence of Pt@Pd/SiO₂ catalyst and Fe²⁺ ion: (a) changes in the UV-Vis absorption spectra of MO during a reaction, (b) effect of total flow rate (N₂/H₂/O₂ mixed gas) on the MO degradation, (c) effect of [Fe²⁺] on the MO degradation, (d) effect of Pt@Pd/SiO₂ catalyst amount on the MO degradation, (e) effect of reaction temperature on the MO degradation, and (f) effect of reaction temperature on the H₂O₂ decomposition and hydrogenation.

3.5. CWPO-II

To explore the possibility of a heterogeneous wastewater treatment method without the addition or generation of undesirable chemicals, a range of bifunctional Fe-M/SiO₂ catalysts were prepared by impregnating Fe on the as-prepared M/SiO₂. PXRD patterns of the Fe-M/SiO₂ catalysts (Fig. S7) showed no significant changes in the crystallinity of Pd

and Pt after Fe impregnation. Moreover, the absence of the characteristic diffraction peak associated with the (110) crystal plane of Fe₂O₃ located at approximately 35.5° suggests that Fe was well dispersed or that the Fe diffraction peak overlapped with that of silica [56].

To investigate the changes in the size and distribution of metal NPs after Fe impregnation, TEM analysis was conducted. As shown in Fig. 12, while there were little changes in the NP sizes of Fe-Pd/SiO₂, Fe-Pt/SiO₂, and Fe-PdPt/SiO₂, the NP sizes of Fe-Pd@Pt/SiO₂ and Fe-Pt@Pd/SiO₂ increased slightly. However, EDS mapping of each catalyst (Fig. S8) indicates that all NPs, including Pd, Pt, and Fe, were well dispersed regardless of the preparation methods.

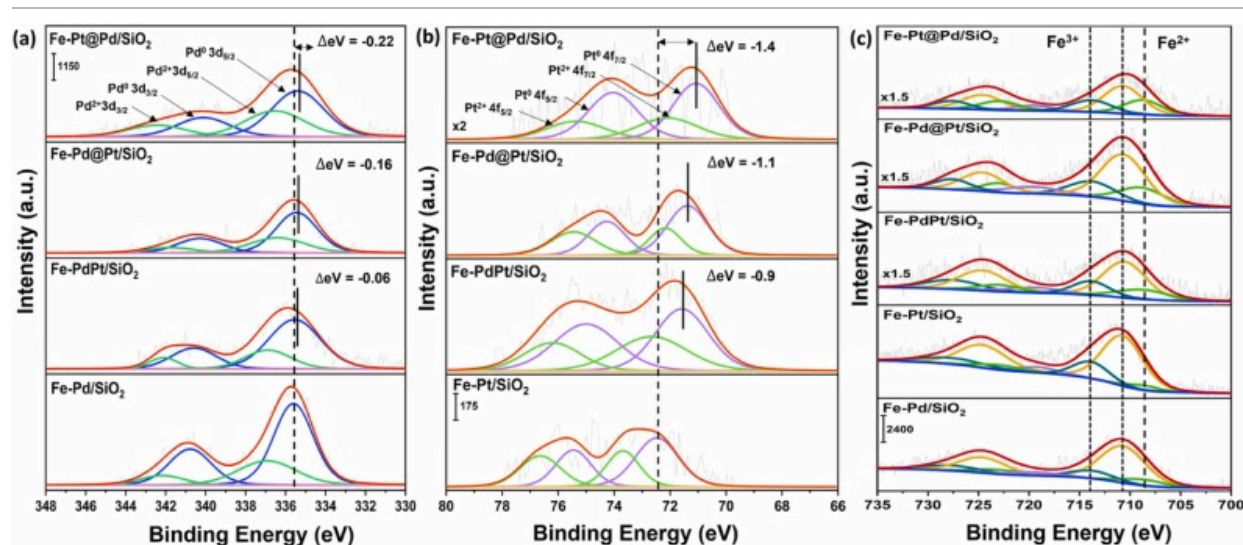


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Fig. 12. TEM analysis of Fe-M/SiO₂ catalysts: (a-e) TEM images, and (f) metal NP sizes based on TEM images.

Fig. 13 shows a series of Pd 3d, Pt 4f, and Fe 2p XPS spectra of Fe-M/SiO₂ catalysts. In the cases of Pd and Pt, not only the deconvoluted peak position of each Pd 3d or Pt 4f orbital, but also the degree of their BE shifts owing to Pd-Pt bimetal formation was almost as same as those of M/SiO₂. The result shows that the impregnation of Fe exerted little impact on the electronic properties of monometallic or bimetallic noble metal NPs. For Fe 2p XPS spectra, the characteristic doublet located at 710.6–713.8 and 724.6–727.6 eV were assigned to the Fe 2p_{1/2} and Fe 2p_{3/2} for the Fe³⁺ species in Fe₂O₃, respectively. The shake-up satellite signal at approximately 719.6 eV was also attributed to the Fe³⁺ species. Moreover, the weak peaks corresponding to the Fe²⁺ species in FeO were also observed at 708.7 (Fe 2p_{1/2}) and 722.9 eV (Fe 2p_{3/2}). The result indicates that both Fe³⁺ (Fe₂O₃) and Fe²⁺ (FeO) species co-exist in the catalysts [57], [58]. However, it was revealed that the distribution of these iron species on a catalyst was significantly affected by alloy formation and preparation method. As shown in Fig. S9, the Fe²⁺/Fe³⁺ ratio of Fe-M/SiO₂ catalyst increased by Pt-Pd alloy formation, and the Fe²⁺/Fe³⁺ ratio of Fe-Pt@Pd/SiO₂ is approximately 4.5 and 2.3 times higher than those of the Fe-Pt/SiO₂ and Fe-Pd/SiO₂, respectively. Since the hydroxy radical is formed by the reaction of Fe²⁺ with H₂O₂ (Fe²⁺ + H₂O₂ → Fe³⁺ + HO· + OH⁻) [59], the high content of Fe²⁺/Fe³⁺ ratio is beneficial in promoting the formation of hydroxy radicals.

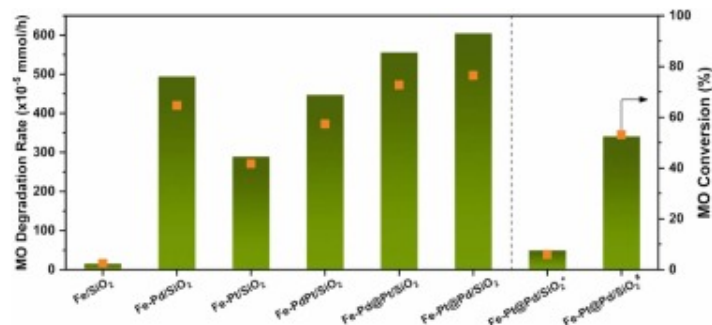


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Fig. 13. (a) Pd 3d, (b) Pt 4 f, and (c) Fe 2p XPS of Fe-M/SiO₂ catalysts.

The reaction performances of the Fe-M/SiO₂ catalysts for the CWPO-II of MO were evaluated, and the results are shown in Fig. 14. The order of catalytic activity among the Fe-M/SiO₂ catalysts for the CWPO-II of MO is exactly matched with that observed for the CWPO-I, which implies again that the efficiency of a catalyst for the *in-situ* H₂O₂ production is of prime importance in attaining a high MO degradation rate. Furthermore, it should be noted that the CWPO-II with *in-situ* generated H₂O₂ was more efficient in MO degradation than that of ready-made H₂O₂.



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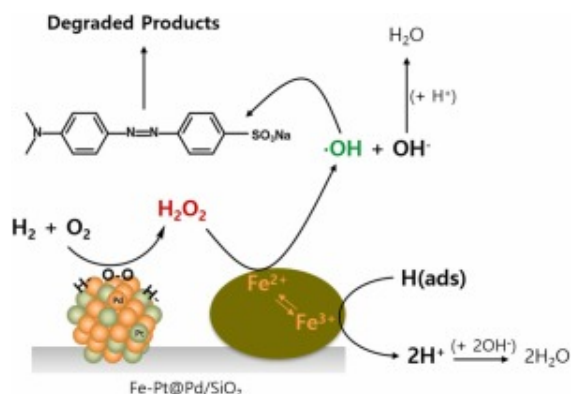
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Fig. 14. CWPO-II of MO with *in-situ* generated H₂O₂. Reaction condition: catalyst (0.1 g), MO (20 ppm), H₂O (150 mL), H₂/O₂/N₂ (mol%) = 3.8/5.0/91.2 (total flow rate = 120 mL/min), 293 K, 1 bar, reaction time = 3 h; * without mixed gas nor H₂O₂; # ready-made H₂O₂ (5 mM) was used instead of mixed gas.

The degree of mineralization of MO during the CWPO-II reaction was investigated by TOC analysis. As summarized in Table S2, the remained organic content after the CWPO-II reaction in the presence of the Fe-Pt@Pd/SiO₂ catalyst was 2.32±0.02 mg/L indicating that the degree of mineralization of MO was approximately 88.4%. It seems that the insufficient mineralization can be ascribed to the incomplete MO conversion during the CWPO-II reaction. Although the reaction solution turned from bright red to transparent after the CWPO-II reaction as shown in Fig. S10, the weak and broad UV-vis absorption band still observed in the solution extracted at the end of reaction suggests that unreacted or partially decomposed organic molecules still remained in the reaction solution.

From a mechanistic aspect, the rate-determining step of the Fenton cycle is the regeneration of the Fe²⁺ from the oxidized Fe³⁺ after the generation of hydroxy radicals [59], [60]. The regeneration of Fe²⁺ from Fe³⁺ within the heterogeneous and mixed Fe₂O₃/FeO domain was relatively slower compared to that of Fe salt in solution, which

might offset the advantage of using the ready-made H₂O₂ in maintaining a high H₂O₂ concentration in solution. In this context, the use of hydrogen during a CWPO-II reaction would accelerate the sluggish regeneration of the iron active site. The chemisorbed H₂ on the surface of Pt@Pd NPs can be readily dissociated into atomic hydrogen that can facilitate the reduction of Fe³⁺ into Fe²⁺, as shown in Fig. 15. A similar beneficial effect of hydrogen on promoting the Fenton cycle was reported for methane oxidation [61].

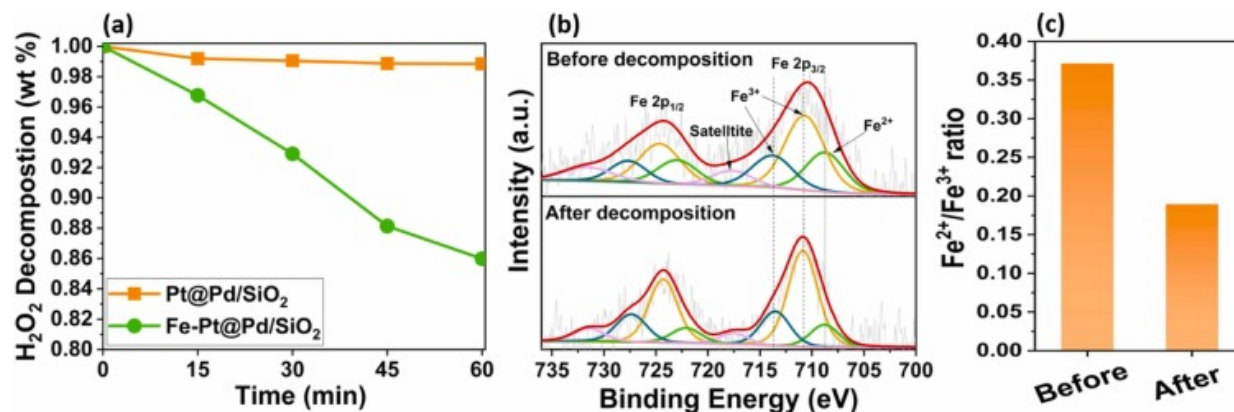


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Fig. 15. Postulated reaction mechanism of the Fe-M/SiO₂-promoted CWPO-II consisting of concerted DSHP and CWPO reactions.

As seen in Fig. 16 (a), the degree of H₂O₂ decomposition by the Fe-Pt@Pd/SiO₂ was compared to that of Pt@Pd/SiO₂. The rapid H₂O₂ decomposition by using the Fe-Pt@Pd/SiO₂ was in contrast with that of Pt@Pd/SiO₂, which strongly suggests that the Fe species on Pt@Pd/SiO₂ is responsible for the Fenton reaction. Moreover, the changes in the Fe 2p XPS (Fig. 16 (b),(c)) of the Pt@Pd/SiO₂ catalyst after H₂O₂ decomposition imply that Fe²⁺ was oxidized to Fe³⁺ after generating hydroxy radicals. The result suggests that the impregnated Fe can efficiently promote the hydroxy radical formation. As shown in Fig. S10, however, while the degree of MO degradation can be improved by increasing the Fe content of Fe-Pt@Pd/SiO₂ up to 2 wt%, it decreased again above 3 wt%.



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Fig. 16. (a) Changes in the H₂O₂ decomposition in the presence of Pt@Pd/SiO₂ or Fe-Pt@Pd/SiO₂, (b) Fe 2p XPS analysis of Fe-Pt@Pd/SiO₂ before and after the H₂O₂ decomposition test, and (c) changes in the Fe²⁺/Fe³⁺ ratio of Fe-Pt@Pd/SiO₂ based on the XPS analysis result.

Based on the discussion above, it is clear that the Fe-Pt@Pd/SiO₂ catalyst is highly efficient in promoting not only the *in-situ* H₂O₂ generation reaction (DSHP), but also the sequential CWPO reaction without any undesirable interference between the two reactions. The unique features of the CWPO reaction with *in-situ* generated H₂O₂ in the presence of Fe-Pt@Pd/SiO₂ catalyst enables the elucidation of a highly efficient and environmentally-benign wastewater treatment method being free from addition and generation of any undesirable chemicals affecting environment.

4. Conclusion

While several efficient wastewater treatment methods belonging to the category of AOPs have been reported, challenging problems persist. These challenges include dosage-dependent efficiency, the generation of various salts, the requirement of an acidic pH

condition, and the need for post-treatment before discharge into the environment. In this context, we envisioned an elegant approach that would enable highly efficient and environmentally-friendly wastewater remediation without the addition or generation of any undesirable chemicals that could harm the environment. To achieve the elusive goal, our aim was to combine the Fe-promoted CWPO reaction of MO with the noble metal-catalyzed DSHP from hydrogen and air, allowing for the *in-situ* generation of H₂O₂ in a concerted manner. To maximize the rate of H₂O₂ production, a new bimetallic Pt@Pd/SiO₂ and Pd@Pt/SiO₂ catalysts were prepared via double-metal complex salt approach, and their reaction performances were evaluated in two CWPO approaches, CWPO-I with soluble Fe salt and CWPO-II with impregnated Fe. It was revealed that the unique characteristics of Pt@Pd NPs, such as a Pd-enriched surface with a strong interaction between Pt and Pd, resulted in superior activity in CWPO-I for MO compared to conventional monometallic (Pd or Pt) or bimetallic PdPt NPs. Furthermore, the excellent activity of the Fe-Pt@Pd/SiO₂ catalyst in CWPO-II with *in-situ* generated H₂O₂ surpassed that achieved with ready-made H₂O₂. This clearly demonstrates the promising potential of the unprecedented approach in realizing highly efficient and eco-friendly wastewater treatment without the addition or generation of any undesirable chemicals that could affect the ecosystem.

CRediT authorship contribution statement

Viet Le Nam Vo: Conceptualization, Methodology, Data curation, Validation, Writing - original draft, **Young-Min Chung:** Conceptualization, Methodology, Resources, Writing - review & editing, Supervision, Project administration, Funding acquisition.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Young-Min Chung reports financial support was provided by National Research

Foundation of Korea. If there are other authors, they declare that they have no known

competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary material

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Supplementary material

Data Availability

No data was used for the research described in the article.

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